NAME

mbm_phins2fnv − Macro to translate processed Phins INS navigation data to an fnv file for use in MB-System processing.

VERSION

Version 5.0

SYNOPSIS

mbm_phins2fnv **−i***inputfile* **−o***outputfile* [**−v** **−h**]

DESCRIPTION

mbm_phins2fnv is a macro intended to translate processed navigation data from an iXblue Phins inertial navigation system (INS) into an "fnv" format navigation file of the kind used elsewhere in **MB-System** processing (see, e.g., the **−N** option of **mbnavlist** and the merge-navigation options of **mbprocess**).

The input file (specified with **−i**/**−−input**) is a processed Phins data file in which each line has the acquisition date and time as a single token (year/month/day followed by a tab and hour:minute:second.fraction), followed by space-separated longitude (degrees), latitude (degrees), heading (degrees), draft (meters), roll (degrees), and pitch (degrees). Lines containing a "#" character (comments) or the word "date" (the column header line) are skipped. Speed and heave are not present in the Phins data and are always written as 0 in the output.

Each remaining line is translated into one line of the output fnv file (specified with **−o**/**−−output**), giving the date and time, the same instant as decimal Unix epoch seconds, longitude, latitude, heading, speed (always 0), draft, roll, pitch, heave (always 0), and four further fields for the port/starboard swath edges (always 0, since the Phins data does not include swath coverage information). A line that cannot be parsed (too few fields, or an unparseable date/time) is skipped with a warning when **−−verbose** is given, and silently otherwise.

Both an input and an output filename must be supplied on the command line; if either is missing the macro prints a diagnostic message and exits without processing.

MB-SYSTEM AUTHORSHIP

David W. Caress
Monterey Bay Aquarium Research Institute
Dale N. Chayes
Center for Coastal and Ocean Mapping
University of New Hampshire
Christian do Santos Ferreira
MARUM - Center for Marine Environmental Sciences
University of Bremen

OPTIONS

−h, **−−help**

This "help" flag causes the macro to print out a description of its command line options and then exit immediately.

−i*inputfile*, **−−input=***inputfile*

Sets the pathname of the input file containing processed Phins INS data. This is a required argument; if it is not supplied the macro prints a message and exits without doing anything further.

−o*outputfile*, **−−output=***outputfile*

Sets the pathname of the output fnv navigation file. This is a required argument; if it is not supplied the macro prints a message and exits without doing anything further.

−v, **−−verbose**
Reports the input and output filenames, warns about (and skips) any input line that cannot be parsed, and prints a final count of lines read, written, and skipped. Without this flag the macro runs silently on success.

SEE ALSO

mbsystem(1), **mbnavlist**(1), **mbprocess**(1)

BUGS

Maybe. Maybe not.